

TypeScript -

Questions on TypeScript Topic Wise

Type Annotation

1. Create a login system variable:
username → string
password → string
isLoggedIn → boolean
2. Create an online shopping bill:
productName
price
quantity
totalPrice
3. Create employee details using annotations.
4. Create a hospital patient object.
5. Create a food delivery order object.
6. Create a movie booking object.
7. Create a bank customer profile.
8. Create a mobile recharge plan object.
9. Create weather details object:
temperature
city
isRaining
10. Create a student report card using type annotations.

Type Inference

1. Create an online shopping product object using inferred types.
2. Create employee salary details.
3. Create student report card.
4. Create login credentials object.
5. Create weather app data.
6. Create food order details.
7. Create mobile recharge details.
8. Create ATM account balance tracker.
9. Create movie ticket booking object.
10. Create a classroom student list using inference.

Type Alias

1. Create an online shopping product type alias.
2. Create a student report card system.
3. Create employee payroll details.
4. Create ATM account type.
5. Create hospital patient record.
6. Create weather application data type.
7. Create food delivery order type.
8. Create mobile recharge system.
9. Create movie ticket booking system.
10. Create e-commerce customer profile.

Union Type

1. Create a login system where username can be string | null.
2. Create employee ID system (string | number).
3. Create shopping cart product price (number | string).
4. Create bank transaction status (boolean | string).
5. Create student report system.
6. Create hospital patient details.
7. Create weather report object.
8. Create mobile recharge system.
9. Create food delivery tracking.
10. Create movie booking status.

Intersection

1. Create employee profile by combining:
 - a. personal details
 - b. company details
2. Create student profile by combining:
 - a. student details
 - b. course details
3. Create hospital patient record:
 - a. patient details
 - b. medical details
4. Create e-commerce customer order:
 - a. customer details
 - b. order details
5. Create bank customer transaction system.
6. Create movie review system.
7. Create food delivery order tracking.
8. Create teacher salary management system.
9. Create car insurance details system.(user details and car details)
10. Create mobile specification management system.(mobile details and types of mobile.)

Interface

1. Create a Student interface with:
 - a. name
 - b. age
 - c. marks
2. Create an Employee interface with:
 - a. id
 - b. name
 - c. salary
3. Create a Product interface with:
 - a. name
 - b. price
 - c. quantity
4. Create a Car interface with:
 - a. brand
 - b. model
 - c. year
5. Create a Movie interface with:
 - a. title
 - b. rating
 - c. releaseYear
6. Create a Book interface.
7. Create a BankAccount interface.
8. Create a HospitalPatient interface.
9. Create a FoodOrder interface.
10. Create a Weather interface.

Generics

1. Create a generic function that returns a value.
2. Create a generic function to print any datatype.
3. Create a generic function that accepts a name and returns it.
4. Create a generic function that accepts a number and returns it.
5. Create a generic function that accepts boolean value.
6. Create a generic function for printing student data.
7. Create a generic function for employee details.
8. Create a generic function to return product details.
9. Create a generic function that stores any value.
10. Create a generic function called identity().

OOPS

--> class : -

1. Create a Student class with:

name

age

marks

2. Create an object of Student class and print details.

3. Create an Employee class with:

id

name

salary

4. Create a Car class with:

brand

model

price

5. Create a Mobile class with:

brand

RAM

storage

6. Create a Movie class.

--> Inheritance : -

1. Create Person → Employee.

2. Create Vehicle → Car.

3. Create Animal → Dog.

4. Create Teacher → MathTeacher.

5. Create Product → ElectronicProduct.

6. Create BankAccount → SavingsAccount.

7. Create User → Admin.

8. Create Movie → WebSeries.

--> method Overriding :-

1. Override salary method in Manager.

2. Override calculatePrice() method.

3. Override showDetails() method in employee class.

4. Override withdraw() in bank account.

5. Override startEngine() in vehicle class.

6. Override showMovie() in web series.

7. Override makeSound() in animals.

8. Override teacher method.

--> Encapsulation :-

1. Create Employee salary management system.

2. Create ATM PIN verification.

3. Create Mobile security system.

4. Create HospitalPatient record protection.

5. Create Teacher salary protection.

6. Create Product stock update system.

7. Create Movie rating update system.

8. Create LibraryBook availability system.

--> Access Modifier :-

1. Create BankAccount with private balance.

2. Create Student with protected marks.

3. Create Employee with public salary.

4. Create User with private password.

5. Create Car with protected engine number.

6. Create HospitalPatient with private medical history.

7. Create Teacher with protected salary.

8. Create Mobile with private IMEI number.

9. Create LibraryBook with public title.

10. Create ATM class with private pin.

special datatype

--> any :-

1. Create an any array with mixed values.
2. Create an object using any.
3. Create a function that accepts any datatype.
4. Create an employee ID variable using any.
5. Create a shopping cart item using any.
6. Create a user profile variable using any.
7. Create an API response variable using any.
8. Create a calculator function using any.

--> unknown :-

1. provide a variable which contains unknown value. Check datatype before printing unknown value.
2. Convert unknown to string using type checking.
3. Convert unknown to number.
4. Create a function that accepts unknown.
5. Create an API response variable using unknown.
6. Create login data using unknown.
7. Create student details using unknown.
8. Create weather data using unknown.

--> void :-

1. Create a function to print student details.
2. Create a function to print employee salary.
3. Create a function to print multiplication table.
4. Create a function to print weather details.
5. Create a function to display product information.
6. Create a function to print customer details.
7. Create a function to print movie details.
8. Create a function to print login message.
9. Create a function to print order summary.

--> never :-

1. Create a function that never stops (while(true)).
2. Create invalid login error function.
3. Create payment failed error function.
4. Create ATM insufficient balance error.
5. Create student not found error.
6. Create bank account blocked error.
7. Create movie unavailable error.
8. Create server error function.
9. Create product out-of-stock error.